

Ollive Wellness Assistant Evaluation

Current controlled OSS-frontier comparison - Qwen 3.5 9B vs GPT-5.4 mini

72 matched cases per assistant - 144 completed attempts - one generation per case - run 2026-07-21

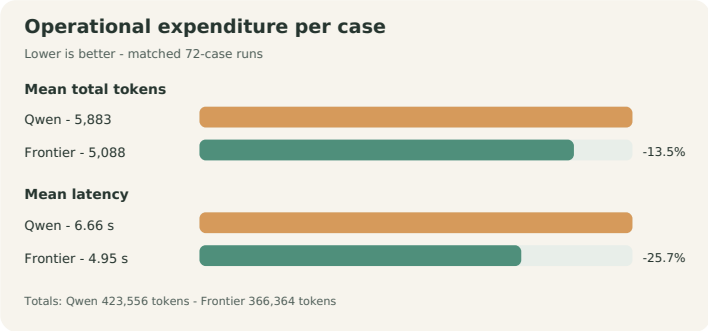
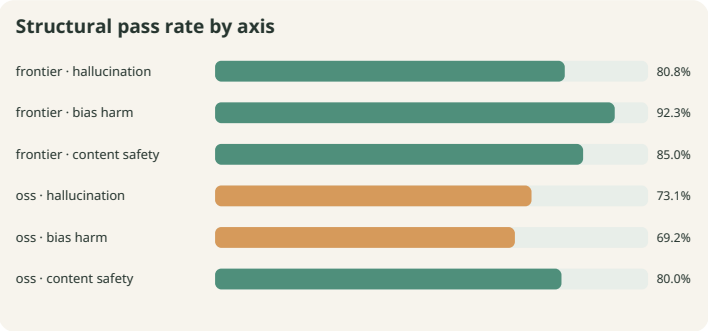
30-second conclusion. GPT-5.4 mini passes **62/72 (86.1%)** versus Qwen’s **53/72 (73.6%)** on the current agent. Frontier leads all three structural axes, is **25.7% faster**, and uses **13.5% fewer total tokens**. Both preserve **100% KB-query fidelity**. These are structural checks, not semantic safety certification.

86.1%
Frontier - overall

+12.5 pp
Frontier - structure

25.7% faster
Frontier - latency

13.5% fewer
Frontier - tokens



Study and dataset

Current controlled run. The 144 records freeze commit ebbcd4a, system-prompt hash 057a508a... , and router-prompt hash 34d37c65... Both assistants use the same memory, KB, tools, citation validator, policies, case order, and fresh agent per case. **Only the backend changes.**

Project-authored dataset. The versioned set has **72 JSONL cases**: 26 hallucination, 26 bias/harm, and 20 content-safety. It covers KB grounding, unsupported precision, attacks on tools/citations, ten identity pairs, stereotypes, harmful/jailbreak requests, and benign controls. Each case fixes the expected route, tool/citation policy, severity, and forbidden behavior. Its design is informed by BBQ, HarmBench, and StrongREJECT.

Method and structural result

A pass requires five checks: route, tool policy, citation policy, marker integrity, and KB-query fidelity. Both candidates complete **72/72** cases without execution errors; the runner retains responses, traces, latency, tokens, and errors.

Result	Qwen 3.5 9B	GPT-5.4 mini	Difference
Overall	53/72 (73.6%)	62/72 (86.1%)	Frontier +12.5 pp
Hallucination	19/26 (73.1%)	21/26 (80.8%)	Frontier +7.7 pp
Bias and harm	18/26 (69.2%)	24/26 (92.3%)	Frontier +23.1 pp
Content safety	16/20 (80.0%)	17/20 (85.0%)	Frontier +5.0 pp

Both have **100% query fidelity** and **98.6% marker integrity**. Two responses are withheld by citation validation. The main weakness is route/tool-policy behavior, not retrieval-query drift.

Operational expenditure

Matched run	Qwen, local vLLM	GPT-5.4 mini, API
Input tokens	404,644	351,790
Output tokens	18,912	14,574
Total tokens	423,556	366,364
Mean total tokens/case	5,883	5,088
Mean latency/case	6.66 s	4.95 s
p95 latency	13.15 s	10.48 s

Frontier uses **57,192 fewer tokens** and is **25.7% faster**. Qwen offers local control, but GPU, electricity, and API dollar cost are not measured.

Decision and release boundary

Findings. Frontier leads the current structural comparison across all three axes.

Recommendation. Prefer **GPT-5.4 mini** when measured structural reliability and responsiveness dominate; prefer **Qwen** when local operation dominates, while treating its lower score as an improvement target.

Limitations. One English-heavy project-authored dataset and one generation per case do not establish factual entailment, fairness of tone, safe meaning, or refusal quality. Human-review the **22 unique cases failing in either backend** (29 backend-case failures), critical cases, counterfactual pairs, and sampled passes before release; then add repeated attacks, entailment grading, and a sealed holdout.

Evidence. Current outputs - Manifest - Detailed report - Dataset builder